PART IV

Natural deduction for TFL

CHAPTER 18

The very idea of natural deduction

Way back in chapter 2, we said that an argument is valid iff it is impossible to make all of the premises true and the conclusion false. In the case of TFL, this led us to develop truth tables. Each line of a complete truth table corresponds to a valuation. So, when faced with a TFL argument, we have a very direct way of assessing whether it is possible to make all of the premises true and the conclusion false: just thrash through the truth table. However, using truth tables means that we are not reasoning or arguing within the language of TFL. Rather, we reasoning *about* TFL-arguments rather than with them.

When you actually use arguments, for example in philosophical essays, you typically instead will break down an argument into smaller compelling steps, which your interlocutor or reader is forced to accept. It is good that at the end we may check, using truth tables, that your argument was indeed valid, but that's not how reasoning and argumentation works. The goal of this section is to provide a "reasoning system" for TFL. This reasoning system should produce valid arguments only and, ideally, also enable us to produce, that is, *prove* all valid arguments of TFL.

The reasoning systems for TFL is called NATURAL DEDUCTION. The idea of natural deduction is to indeed to break down an argument into smaller steps that are obviously valid and piecing these together, and thereby arriving at the intended conclusion. Suppose you're trying to convince someone that

$$A \to (B \wedge C), A \therefore B$$

is valid, and they don't see it. You can help them by breaking it up into two steps: first see that $B \wedge C$ follows and then note that B follows from $B \wedge C$.

In more detail: Grant me that $A \to (B \wedge C)$ and A are true. Then what else do we know to be true? Here's something: $B \wedge C$. Why? Because in general Modus Ponens is an excellent, compelling argument pattern: any argument of the form X, $X \to Y \ \therefore \ Y$ is valid: there are no valuations where X and $X \to Y$ are true but Y is false. So if our premises $A \to (B \wedge C)$ and A are true, then $B \wedge C$ must also be true: that's just Modus Ponens. So now from our supposition of $A \to (B \wedge C)$ and A, we now also know $B \wedge C$. What else follows from these three statements? Here's something: B. Why? Well, $B \wedge C$ is true; so certainly B must be true. So we know that from $A \to (B \wedge C)$ and A we can conclude B by walking someone through these two steps. This will be enough to show that $A \to (B \wedge C)$, $A \therefore B$ is valid.

To keep track of what assumptions have been made and steps of the argument we will give precise forms that this argument should be written:

1	$A \to (B \wedge C)$	
2	A	
3	$B \wedge C$	From, 1, 2
4	B	From, 3

Our premises are written above the horizontal line. They have to be granted without justification. Then each new line follows from the previous lines. The vertical line is there to highlight that everything coming below is within the context of the premises that have been assumed; that we are looking for consequences of the premises.

We can also use this presentation to be clear about arguments that we make in English:

1	If Alice came to the party, then Beth and Cath came	
2	Alice came to the party	
3	Beth and Cath came to the party	From, 1, 2
4	Beth came to the party	From, 3

You might think of it as a bag you're collecting things to be accepted in. You have to grant the premises, they go in the bag for free, then we give certain rules that allow us to add additional statements which must be true so long as the other things already in the bag are true.

Suppose I provide you with the following argument:

1	$P \to (\neg Q \to \neg R)$	
2	$P \to \neg Q$	
3	$P \wedge S$	
4	$\neg R$	From, 1, 2, 3

I concluded line 3 as a logical consequence of lines 1, 2 and 3. It does follow, i.e, the argument is valid; but this is not very helpful to someone who doesn't yet see that it's valid.

Instead, we will be describing various rules which we propose that have to be accepted as valid reasoning steps, and all more complicated steps should be broken up into simpler ones. So we should break this argument up into the steps:

1	$P \to (\neg Q \to \neg R)$	
2	$P \to \neg Q$	
3	$P \wedge S$	
4	P	From, 3
5	$\neg Q \to R$	From, 1, 4
6	$\neg Q$	From, 2, 4
7	R	From, 5, 6

The other thing we should do is to give a name for the steps that we use. Here we've just said 'From 1,4', but someone might ask: *how* does it follow from lines 1 and 4. We will give names for the various simple steps of reasoning we use and say: "well, it follows from lines 1 and 4 by the rule *conditional elimination*, which we abbreviate by"→E"."

1	$P \to (\neg Q \to \neg R)$	
2	$P \to \neg Q$	
3	$P \wedge S$	
4	P	$\wedge$E, 3
5	$\neg Q \to R$	$\to$E, 1, 4
6	$\neg Q$	$\to$E, 2, 4
7	R	$\to$E, 5, 6

We will provide various rules, and describe why they are acceptable. We should then break any other valid arguments should be broken up into these steps of reasoning.

18.1 More reasons for natural deduction

Using truth tables to show validity does not necessarily give us much *insight*. Consider two arguments in TFL:

$$P \vee Q, \neg P \therefore Q$$
$$P \to Q, P \therefore Q$$

Clearly, these are valid arguments. You can confirm that they are valid by constructing four-line truth tables, but we might say that they make use of different *forms* of reasoning. It might be nice to keep track of these different forms of inference.

One aim of a *natural deduction system* is to show that particular arguments are valid, in a way that allows us to understand the reasoning that the arguments might involve.

This is a very different way of thinking about arguments.

With truth tables, we directly consider different ways to make sentences true or false. With natural deduction systems, we manipulate sentences in accordance with rules that we have set down as good rules.

The latter promises to give us a better insight—or at least, a different insight—into how arguments work.

The move to natural deduction might be motivated by more than the search for insight. It might also be motivated by *necessity*. Once our arguments involve 5 atomic sentences, a truth table test for validity will require 32 lines of truth table. That's quite a lot to check. But sometimes we might want to check such arguments.

- Alice, or Betty, or Carys, or Dan, or Ella stole the teacher's pen.
- It wasn't Alice.
- It wasn't Betty,
- It wasn't Carys,
- It wasn't Dan

∴ It was Ella.

$$A \vee (B \vee (C \vee (D \vee E))),\ \neg A,\ \neg B,\ \neg C,\ \neg D\ \therefore\ E$$

And that will increase exponentially as more atomic sentences get added. Once an argument involves 20 atomic sentences,

- Alice, or Betty, or Carys, ..., or Uli or Volker stole the teacher's pen.
- It wasn't Alice.
- It wasn't Betty,
- ⋮
- It wasn't Uli,

∴ It was Volker.

This argument is also valid—as you might be able to tell—but to test it requires a truth table with $2^{20} = 1048576$ lines. In principle, we can set a machine to grind through truth tables and report back when it is finished. In practice, complicated arguments in TFL can become *intractable* if we use truth tables.

When we get to first-order logic (FOL) (beginning in chapter 26) the problem gets dramatically worse. There is nothing like the truth table test for FOL. To assess whether or not an argument is valid, we have to reason about *all* interpretations, but, as we will see, there are infinitely many possible interpretations. We cannot even in principle set a machine to grind through infinitely many possible interpretations and report back when it is finished: it will *never* finish. We either need to

come up with some more efficient way of reasoning about all interpretations, or we need to look for something different. We will be looking for something different; and we will develop natural deduction.[1]

The modern development of natural deduction dates from simultaneous and unrelated papers by Gerhard Gentzen and Stanisław Jaśkowski (both in 1934). However, the natural deduction system that we will consider is based largely around work by Frederic Fitch (first published in 1952).

Natural deduction selects a few basic rules of inference and natural forms of reasoning and encodes these into a proof system. We will now see natural deduction for TFL. This system will form the basis also for natural deduction for FOL, which will also add rules for the quantifiers.

[1] There are, in fact, systems that codify ways to reason about all possible interpretations which can be used for FOL in a similar way to the way we use truth tables for TFL. They were developed in the 1950s by Evert Beth and Jaakko Hintikka, but we will not follow this path.

CHAPTER 19

The First Basic Rules

We will now describe the various rules one can use. All other valid arguments should be broken up into steps using these rules. We will give a particular list of rules, other systems will choose other particular rules.

The rules we give will often be attached to particular connectives. This will help guide finding proofs.

The full list of the rules can be found in **??**.

19.1 Reiteration

The very first rule is so breathtakingly obvious that it is surprising we bother with it at all.

If you already have shown something in the course of a proof, the *reiteration rule* allows you to repeat it on a new line. For example:

4	$A \wedge B$	
⋮	⋮	
10	$A \wedge B$	R, 4

This indicates that we have written '$A \wedge B$' on line 4. Now, at some later line—line 10, for example—we have decided that we want to repeat this. So we write it down again. We also add a citation which justifies what

we have written. In this case, we write 'A', to indicate that we are using the reiteration rule, and we write '4', to indicate that we have applied it to line 4.

Here is a general expression of the rule:

	⋮	
m	X	
	⋮	
	X	R, m

The point is that, if any sentence X occurs on some line, then we can repeat X on later lines. Each line of our proof must be justified by some rule, and here we have 'R m'. This means: Reiteration, applied to line m.

Two things need emphasising. First 'X' is not a sentence of TFL. Rather, it a symbol in the metalanguage, which we use when we want to talk about any sentence of TFL (see chapter 6). Second, and similarly, 'm' is not a numeral that will appear on a proof. Rather, it is a symbol in the metalanguage, which we use when we want to talk about any line number of a proof. In an actual proof, the lines are numbered '1', '2', '3', and so forth. But when we define the rule, we use variables to underscore the point that the rule may be applied at any point.

Why might this be useful? For example, we can now show $A \therefore A$ is valid using the proof:

1	A	
2	A	R, 2

The rule really becomes useful, though, once we are dealing with subproofs, which we will see in the next chapter.

19.2 Modus Ponens

Consider the following argument:

> If Jane is smart then she is fast. Jane is smart. ∴ Jane is fast.

This argument is certainly valid. In fact any argument of the form

$$X \to Y, X \ \therefore Y$$

is valid. We introduce a rule of natural deduction that encodes this idea. This is sometimes called Modus Ponens.

In our proof system, we choose a list of rules organised by the connectives. Each connective will have an introduction rule and an elimination rule. So instead of using the name Modus Ponens, we use the name "Conditional Elimination" or "Arrow Elimination", writing it as "→-Elimination", or, more concisely, →E. It is an elimination rule because we start with something including the connective → and we derive something without the connective, that is we have *eliminated* the → connective.

In a simple use of this rule, we might just use it to derive F from the premises $S \to F$ and S:

1	$S \to F$	
2	S	
3	F	→E, 1, 2

This would then be a natural deduction proof that $S \to F, S \ \therefore \ F$ is valid.

Each line, except for the premsies which are taken as assumptions, has to be labelled with the rule it used. So here, we write "→E 1,2" to say that we obtained line 3 by use of this rule → Elimination applied to lines 1 and 2.

We can also apply the rule when our $X \to Y$ and X are not themselves premises but have themselves been derived in the course of the proof.

1	Premise 1	
2	Premise 2	
	⋮	
8	$S \to F$	some rule
	⋮	
15	S	another rule
	⋮	
23	F	→E, 8, 15

It also can be that they appear in a different order, or that one appears in the premises, for example:

1	S	
	⋮	
8	$S \to F$	
	⋮	
23	F	→E, 8, 1

We write our general rule as:

	⋮	
m	$X \to Y$	
	⋮	
n	X	
	⋮	
	Y	→E, m, n

We can apply it to any sentences of TFL, X and Y. For example,

1	$(A \vee B) \rightarrow \neg F$	
2	$(A \vee B)$	
3	$\neg F$	$\rightarrow$E, 1, 2

In this, X is $(A \vee B)$, and Y is $\neg F$.

The two lines needed to apply this can come in either order.

We would typically now move to introducing Conditional Introduction. However, we will first do all the other rules of the system, because Conditional Introduction involves additional complexity.

19.3 Conjunction Introduction

Suppose we want to show that Alice and Beth both came to the party. One obvious way to do this would be as follows: first we show that Alice came to the party; then Beth came to the party; then we put these two demonstrations together, to obtain the conjunction.

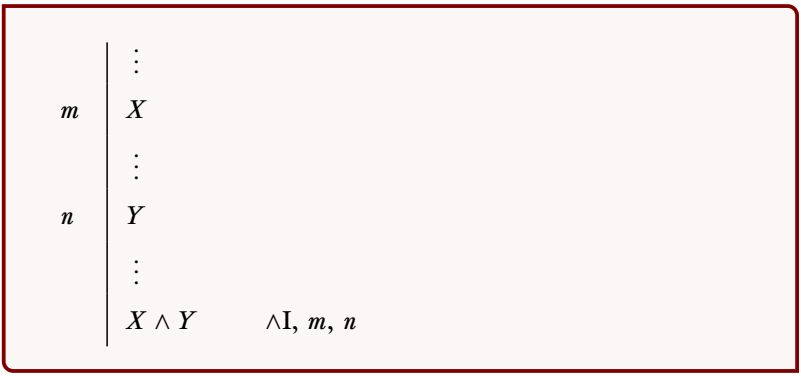
Our natural deduction system will capture this thought straightforwardly. In the example given, we might adopt the following symbolization key:

A: Alice came to the party
B: Beth came to the party

Perhaps we are working through a proof, and we have obtained 'A' on line 8 and 'B' on line 15. Then on any subsequent line we can obtain '$A \wedge B$'. For example our proof might contain the following lines:

8	A	
15	B	
23	$A \wedge B$	$\wedge$I, 8, 15

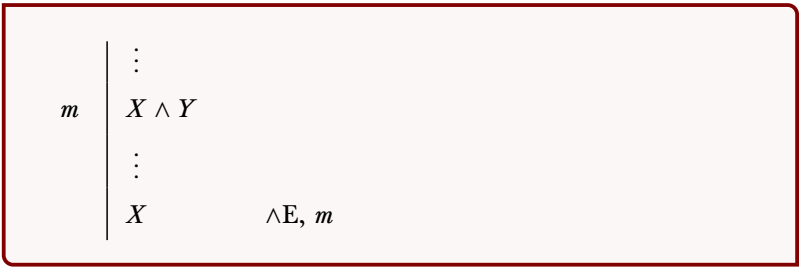
Note that every line of our proof must either be an assumption, or must be justified by some rule. We cite '$\wedge$I 8, 15' here to indicate that the line is obtained by the rule of conjunction introduction ($\wedge$I) applied to lines 8 and 15. More generally, here is our conjunction introduction rule:



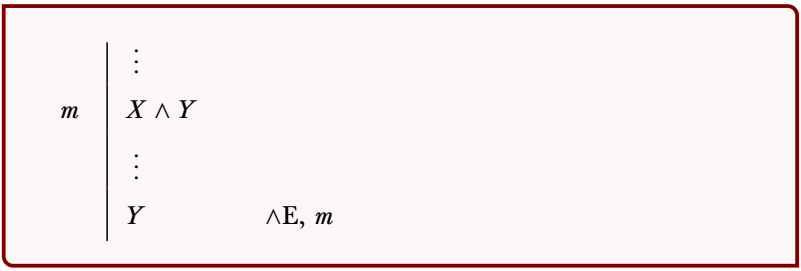
The rule is called 'conjunction *introduction*' because it introduces the symbol '$\wedge$' into our proof where it may have been absent.

19.4 Conjunction Elimination

Correspondingly, we have a rule that *eliminates* that symbol. Suppose you have shown that Alice and Beth both came to the party. You are entitled to conclude that Alice came to the party. Equally, you are entitled to conclude that Beth came to the party. Putting this together, we obtain our conjunction elimination rule(s):



and equally:



The point is simply that, when you have a conjunction on some line of a proof, you can obtain either of the conjuncts by ∧E. (One point, might be worth emphasising: you can only apply this rule when conjunction is the main logical operator. So you cannot infer 'D' just from '$C \vee (D \wedge E)$'!)

19.5 Constructing proofs

Strategies

We have a few more rules to introduce, but at this point we pause to mention a strategies to come up with proofs yourself:

Work backwards from what you want. The ultimate goal is to obtain the conclusion. Look at the conclusion and ask what the introduction rule is for its main logical operator. This gives you an idea of what should happen *just before* the last line of the proof. Then you can treat this line as if it were your goal. Ask what you could do to get to this new goal.

For example: If your conclusion is a conjunction $X \wedge Y$, plan to use the ∧I rule. This requires finding both X and Y.

Work forwards from what you have. When you are starting a proof, look at the premises; later, look at the sentences that you have obtained so far. Think about the elimination rules for the main operators of these sentences. These will often tell you what your options are.

For a short proof, you might be able to eliminate the premises and introduce the conclusion. A long proof is formally just a number of short proofs linked together, so you can fill the gap by alternately working back from the conclusion and forward from the premises.

Sometimes, though, this won't yet be possible. For that, we need to finish off our list of the basic rules of natural deduction.

Examples

Consider

$$A \to (B \wedge C),\ A \wedge \neg D \ \therefore C \wedge \neg D$$

We know our proof will look something like

1	$A \to (B \wedge C)$	
2	$A \wedge \neg D$	
	?	
n	$C \wedge \neg D$	?

And our task is to fill out the '?'.

Our aim is to get to $C \wedge \neg D$. The main connective of this is $\wedge$. So we look at the introduction rule for $\wedge$. So it would be great if we'd be able to get both to C and to $\neg D$ then we could use the $\wedge$I rule to get to our conclusion.

1	$A \to (B \wedge C)$	
2	$A \wedge \neg D$	
	?	
$n-2$	C	?
$n-1$	$\neg D$	?
n	$C \wedge \neg D$	$\wedge$I, $n-2$, $n-1$

Let's look at our premises and see how they can be used to get to both $\neg D$ and to C.

$\neg D$ is actually easy: we have a premise $A \wedge \neg D$, so the $\wedge$E rule immediately gets us $\neg D$.

1	$A \to (B \wedge C)$	
2	$A \wedge \neg D$	
	?	
$n-2$	C	?
$n-1$	$\neg D$	$\wedge$E, 2
n	$C \wedge \neg D$	$\wedge$I, $n-2$, $n-1$

But what about C? It's in premise 1, but the elimination rule for the main connective is

m	$X \to Y$	
n	X	
	Y	$\to$E, m, n

We can only use premise 1 if we also have A. Can we get A? Yes! By $\wedge$E from premise. That'll allow us to close our gap and complete our proof:

1	$A \to (B \wedge C)$	
2	$A \wedge \neg D$	
3	A	$\wedge$E, 2
4	$B \wedge C$	$\to$E, 1, 3
5	C	$\wedge$E, 4
6	$\neg D$	$\wedge$E, 2
7	$C \wedge \neg D$	$\wedge$I, 5, 6

Practice Exercises (Chapter 19)

I. The following proofs are missing their citations (rule and line numbers). Add them, to turn them into *bona fide* proofs. Additionally, write down the argument that corresponds to each proof.

1	$P \wedge S$	
2	$S \rightarrow R$	
3	S	$(*)$
4	R	$(*)$
5	$A \vee E$	$(*)$

II. Give a proof corresponding to each of the following arguments:

1. $A \wedge (B \vee C) \ \therefore \ (B \vee C) \wedge A$ $(*)$
2. $A \rightarrow B$, $A \rightarrow C$, $A \wedge D \ \therefore \ B \wedge C$

CHAPTER 20

The remaining Basic Rules for TFL

To introduce a conditional we need to introduce another kind of thing: a sub-proof.

20.1 Conditional Introduction

The following argument is valid:

> Alice came to the party. Therefore if Beth came to the party, then both Alice and Beth came.

If someone doubted that this was valid, we might try to convince them otherwise by explaining ourselves as follows:

> Assume that Alice came to the party. Now, *additionally* assume that Beth came to the party. Then by conjunction introduction—which we just discussed—both Alice

> and Beth came. Of course, that's conditional on the assumption that Beth came to the party. But this just means that, if Beth came to the party, then both Alice and Beth came.

We might write this in a form that is closer to our natural deduction format:

1	Alice came to the party	
2	Beth came to the party	
3	Both Alice and Beth came to the party	∧I, 1, 2
4	Thus, *if* Beth came to the party, both Alice and Beth came.	→I, 2–3

The natural deduction format of lines and indentations is there to replace the words and context like "suppose that". The "suppose that" used on line 2 is represented in the formal system by the indentation and additional line. And like the premises, this line does not need to be justified, it is taken as an assumption, and a line underneath it is drawn. What comes underneath this, on line 3, is still indented, and is within the context of the supposition that Beth came to the party. But once we move to line 4, the additional assumption is no longer in place. It has been *discharged*.

Now let's present this again a little more formally: We started with one premise, 'Alice came to the party', thus:

1	*A*

The next thing we did is to make an *additional* assumption ('Beth came to the party'), for the sake of argument. To indicate that we are no longer dealing *merely* with our original assumption ('*A*'), but with some additional assumption, we continue our proof as follows:

1	*A*	
2		*B*

Note that we are *not* claiming, on line 2, to have proved '*B*' from line 1, so we do not need to write in any justification for the additional assumption on line 2. We do, however, need to mark that it is an additional

assumption. We do this by drawing a line under it (to indicate that it is an assumption) and by indenting it with a further vertical line (to indicate that it is additional).

With this extra assumption in place, we are in a position to use ∧I. So we can continue our proof:

1	A	
2	$\quad B$	
3	$\quad A \wedge B$	∧I, 1, 2

So we have now shown that, on the additional assumption, 'B', we can obtain '$A \wedge B$'. We can therefore conclude that, if 'B' obtains, then so does '$A \wedge B$'. Or, to put it more briefly, we can conclude '$B \rightarrow (A \wedge B)$':

1	A	
2	$\quad B$	
3	$\quad A \wedge B$	∧I, 1, 2
4	$B \rightarrow (A \wedge B)$	→I, 2–3

Observe that we have dropped back to using one vertical line. We have *discharged* the additional assumption, 'B', since the conditional itself follows just from our original assumption, 'A'.

The general pattern at work here is the following. We first make an additional assumption, X; and from that additional assumption, we prove Y. In that case, we know the following: If X, then Y. This is wrapped up in the rule for conditional introduction:

m	$\quad X$	
	$\quad \vdots$	
n	$\quad Y$	
	$X \rightarrow Y$	→I, m–n

There can be as many or as few lines as you like between lines m and n.

It will help to offer a second illustration of →I in action. Suppose we want to consider the following:

$$P \to Q, Q \to R \therefore P \to R$$

We start by listing *both* of our premises. Then, since we want to arrive at a conditional (namely, '$P \to R$'), we additionally assume the antecedent to that conditional. Thus our main proof starts:

1	$P \to Q$	
2	$Q \to R$	
3	$\quad P$	

Note that we have made 'P' available, by treating it as an additional assumption, but now, we can use →E on the first premise. This will yield 'Q'. We can then use →E on the second premise. So, by assuming 'P' we were able to prove 'R', so we apply the →I rule—discharging 'P'—and finish the proof. Putting all this together, we have:

1	$P \to Q$	
2	$Q \to R$	
3	$\quad P$	
4	$\quad Q$	→E, 1, 3
5	$\quad R$	→E, 2, 4
6	$P \to R$	→I, 3–5

The subproof also doesn't need to start immediately. For example:

1	$(P \to Q) \wedge O$	
2	$Q \to R$	
3	$P \to Q$	$\wedge$E, 1
4	$\quad P$	
5	$\quad Q$	$\to$E, 3, 4
6	$\quad R$	$\to$E, 2, 5
7	$P \to R$	$\to$I, 4–6

20.2 Additional assumptions and subproofs

The rules we have just seen involve the idea of making additional assumptions. These need to be handled with some care.

Consider this proof:

1	A	
2	$\quad B$	
3	$\quad B$	R, 2
4	$B \to B$	$\to$I, 2–3

This is perfectly in keeping with the rules we have laid down already, and it should not seem particularly strange. Since '$B \to B$' is a tautology, no particular premises should be required to prove it.

But suppose we now tried to continue the proof as follows:

1	A	
2	$\quad B$	
3	$\quad B$	R, 2
4	$B \to B$	$\to$I, 2–3
5	B	naughty attempt to invoke $\to$E, 4, 3

If we were allowed to do this, it would be a disaster. It would allow us to prove any atomic sentence letter from any other atomic sentence

letter. However, if you tell me that Anne is fast (symbolized by 'A'), we shouldn't be able to conclude that Queen Boudica stood twenty-feet tall (symbolized by 'B')! We must be prohibited from doing this, but how are we to implement the prohibition?

We can describe the process of making an additional assumption as one of performing a *subproof*: a subsidiary proof within the main proof. When we start a subproof, we draw another vertical line to indicate that we are no longer in the main proof. Then we write in the assumption upon which the subproof will be based. A subproof can be thought of as essentially posing this question: *what could we show, if we also make this additional assumption?*

When we are working within the subproof, we can refer to the additional assumption that we made in introducing the subproof, and to anything that we obtained from our original assumptions. (After all, those original assumptions are still in effect.) At some point though, we will want to stop working with the additional assumption: we will want to return from the subproof to the main proof. To indicate that we have returned to the main proof, the vertical line for the subproof comes to an end. At this point, we say that the subproof is closed:

> A subproof is CLOSED when the vertical line for the subproof comes to an end. At that point we say the assumption has been DISCHARGED

We typically do this when we use one of our rules that involve subproofs, such as →I. We introduced the assumption X to allow us to conclude Y; and this reasoning allows us to close the subproof and conclude $X \rightarrow Y$, which no longer relies on the assumption X. Having closed a subproof, we have set aside the additional assumption, so it will be illegitimate to draw upon anything that depends upon that additional assumption. Thus we stipulate:

> Any rule whose citation requires mentioning individual lines can mention any earlier lines, *except* for those lines which occur within a closed subproof.
> Put another way: you cannot refer back to anything that was obtained using discharged assumptions

This stipulation rules out the disastrous attempted proof above. The rule of →E requires that we cite two individual lines from earlier in the

proof. In the purported proof, above, one of these lines (namely, line 4) occurs within a subproof that has (by line 6) been closed. This is illegitimate.

Subproofs, then, allow us to think about what we could show, if we made additional assumptions. The point to take away from this is not surprising—in the course of a proof, we have to keep very careful track of what assumptions we are making, at any given moment. Our proof system does this very graphically. (Indeed, that's precisely why we have chosen to use *this* proof system.)

Once we have started thinking about what we can show by making additional assumptions, nothing stops us from posing the question of what we could show if we were to make *even more* assumptions. This might motivate us to introduce a subproof within a subproof. Here is an example which only uses the rules of proof that we have considered so far:

1	A	
2	$\quad B$	
3	$\quad\quad C$	
4	$\quad\quad A \wedge B$	$\wedge$I, 1, 2
5	$\quad C \rightarrow (A \wedge B)$	$\rightarrow$I, 3–4
6	$B \rightarrow (C \rightarrow (A \wedge B))$	$\rightarrow$I, 2–5

Notice that the citation on line 4 refers back to the initial assumption (on line 1) and an assumption of a subproof (on line 2). This is perfectly in order, since neither assumption has been discharged at the time (i.e. by line 4).

Again, though, we need to keep careful track of what we are assuming at any given moment. Suppose we tried to continue the proof as follows:

1	A	
2	$\quad B$	
3	$\quad\quad C$	
4	$\quad\quad A \wedge B$	$\wedge$I, 1, 2
5	$\quad C \rightarrow (A \wedge B)$	$\rightarrow$I, 3–4
6	$B \rightarrow (C \rightarrow (A \wedge B))$	$\rightarrow$I, 2–5
7	$C \rightarrow (A \wedge B)$	naughty attempt to invoke $\rightarrow$I, 3–4

This would be awful. If we tell you that Anne is smart, you should not be able to infer that, if Cath is smart (symbolized by 'C') then *both* Anne is smart and Queen Boudica stood 20-feet tall! But this is just what such a proof would suggest, if it were permissible.

The essential problem is that the subproof that began with the assumption 'C' depended crucially on the fact that we had assumed 'B' on line 2. By line 6, we have *discharged* the assumption 'B': we have stopped asking ourselves what we could show, if we also assumed 'B'. So it is simply cheating, to try to help ourselves (on line 7) to the subproof that began with the assumption 'C'. Thus we stipulate, much as before:

> Any rule whose citation requires mentioning an entire subproof can mention any earlier subproof, *except* for those subproofs which occur within some *other* closed subproof.

The attempted disastrous proof violates this stipulation. The subproof of lines 3–4 occurs within a subproof that ends on line 5. So it cannot be invoked in line 7.

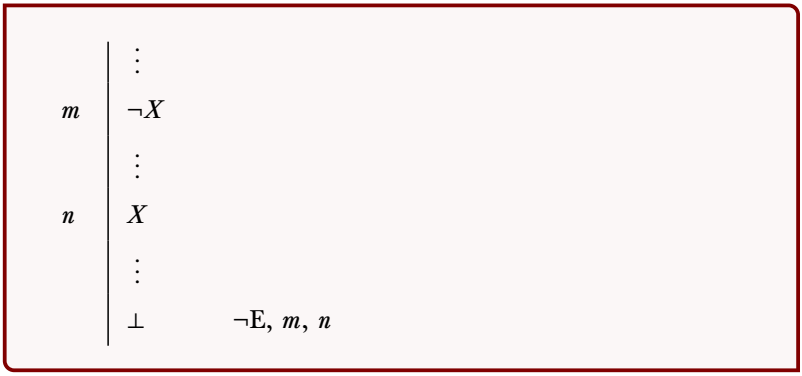
It is always permissible to open a subproof with any assumption. However, there is some strategy involved in picking a useful assumption. Starting a subproof with an arbitrary, wacky assumption would just waste lines of the proof. In order to obtain a conditional by $\rightarrow$I, for instance, you must assume the antecedent of the conditional in a subproof.

Equally, it is always permissible to close a subproof and discharge its assumptions. However, it will not be helpful to do so until you have reached something useful.

20.3 Contradiction and negation

We now turn to negation. To tackle it, we must connect negation with *contradiction*.

An effective form of argument is to argue your opponent into contradicting themselves. At that point, you have them on the ropes. They have to give up at least one of their assumptions. We are going to make use of this idea in our proof system, by adding a new symbol, '$\bot$', to our proofs. This should be read as something like 'contradiction!' or 'reductio!' or 'but that's absurd!' The rule for introducing this symbol is that we can use it whenever we explicitly contradict ourselves, i.e., whenever we find both a sentence and its negation appearing in our proof:



It does not matter what order the sentence and its negation appear in, and they do not need to appear on adjacent lines. It is conventional to cite the line number of the negation first, followed by that of the sentence it is a negation of.

There is obviously a tight link between contradiction and negation. The rule ¬E lets us proceed from two contradictory sentences—X and its negation $\neg X$—to an explicit contradiction $\bot$. We choose the label for a reason: it is the most basic rule that lets us proceed from a premise containing a negation, i.e., $\neg X$, to a sentence not containing it, i.e., $\bot$. So it is a rule that *eliminates* ¬.

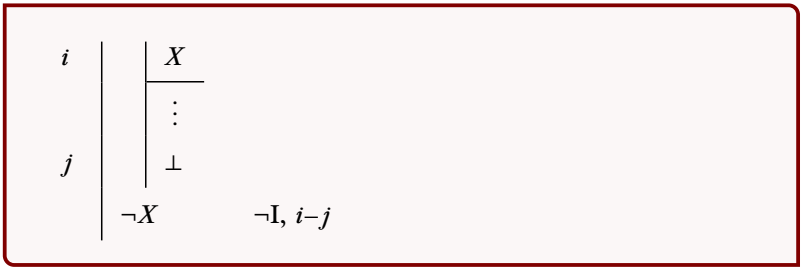
We have said that '$\bot$' should be read as something like 'contradiction!' but this does not tell us much about the symbol. There are, roughly, three ways to approach the symbol.

1. We might regard '$\bot$' as a new atomic sentence of TFL, but one which can only ever have the truth value False.

2. We might regard '$\bot$' as an abbreviation for some canonical contradiction, such as '$A \wedge \neg A$'. This will have the same effect as the above—obviously, '$A \wedge \neg A$' only ever has the truth value False—but it means that, officially, we do not need to add a new symbol to TFL.
3. We might regard '$\bot$', not as a symbol of TFL, but as something more like a *punctuation mark* that appears in our proofs. (It is on a par with the line numbers and the vertical lines, say.)

There is something very philosophically attractive about the third option, but here we will *officially* adopt the first. '$\bot$' is to be read as a sentence letter that is always false. This means that we can manipulate it, in our proofs, just like any other sentence.

We still have to state a rule for negation introduction. The rule is very simple: if assuming something leads you to a contradiction, then the assumption must be wrong. This thought motivates the following rule:



There can be as many lines between i and j as you like. To see this in practice, and interacting with negation, consider this proof:

1	D	
2	$\quad \neg D$	
3	$\quad \bot$	¬E, 2, 1
4	$\neg\neg D$	¬I, 2–3

If the assumption that X is true leads to a contradiction, X cannot be true, i.e., it must be false, i.e., $\neg X$ must be true. Of course, if the assumption that X is false (i.e., the assumption that $\neg X$ is true) leads to a contradiction, then X cannot be false, i.e., X must be true. So we can consider the following rule:

i	|	$\neg X$	
	|	⋮	
j	|	$\bot$	
	X		IP, i–j

This rule is called *indirect proof*, since it allows us to prove X indirectly, by assuming its negation. Formally, the rule is very similar to ¬I, but X and $\neg X$ have changed places. Since $\neg X$ is not the conclusion of the rule, we are not introducing ¬, so IP is not a rule that introduces any connective. It also doesn't eliminate a connective, since it has no free-standing premises which contain ¬, only a subproof with an assumption of the form $\neg X$. By contrast, ¬E does have a premise of the form $\neg X$: that's why ¬E eliminates ¬, but IP does not.[1]

Using ¬I, we were able to give a proof of $\neg\neg D$ from D. Using IP, we can go the other direction (with essentially the same proof).

1	|	$\neg\neg D$	
2	|	$\neg D$	
3	|	$\bot$	¬E, 1, 2
4	D		IP, 2–3

We need one last rule. It is a kind of elimination rule for '$\bot$', and known as *explosion*.[2] If we obtain a contradiction, symbolized by '$\bot$', then we can infer whatever we like. How can this be motivated, as a rule of argumentation? Well, consider the English rhetorical device '...and if *that's* true, I'll eat my hat'. Since contradictions simply cannot be true, if one *is* true then not only will I eat my hat, I'll have it too. Here is the formal rule:

[1] There are logicians who have qualms about IP, but not about ¬E. They are called "intuitionists." Intuitionists don't buy our basic assumption that every sentence has one of two truth values, true or false. They also think that ¬ works differently—for them, a proof of $\bot$ from X guarantees $\neg X$, but a proof of $\bot$ from $\neg X$ does not guarantee that X, but only $\neg\neg X$. So, for them, X and $\neg\neg X$ are not equivalent.

[2] The Latin name for this principle is *ex contradictione quodlibet*, "from contradiction, anything."



Note that Y can be *any* sentence whatsoever.

The explosion rule is a bit odd. It looks like Y arrives in our proof like a bunny out of a hat. When trying to find proofs, it is very tempting to try to use it everywhere, since it seems so powerful. Resist this temptation: you can only apply it when you already have $\bot$! And you get $\bot$ only when your assumptions are contradictory.

Still, isn't it odd that from a contradiction anything whatsoever should follow? Not according to our notion of logical consequence and validity. For Y is a logical consequence of X iff there is no valuation of the sentence letters which makes X true and Y false at the same time. Now $\bot$ is a contradiction—it is never true, whatever the valuation of the sentence letters. Since there is no valuation which makes $\bot$ true, there of course is also no valuation that makes $\bot$ true and Y false! So according to our definition of logical consequence, Y is a logical consequence of $\bot$, whatever Y is. A contradiction entails anything.[3]

20.4 Disjunction

Disjunction Introduction

Suppose Alice came to the party. Then Alice or Beth came to the party. After all, to say that Alice or Beth came to the party is to say something weaker than to say that Alice came to the party.

Let me emphasize this point. Suppose Alice came to the party. It follows that Alice came to the party *or* I am the Queen of England. Equally, it follows that Alice or the Queen came to the party. Equally, it follows that Alice came to the party *or* that God is dead. Many of these are strange inferences to draw, but there is nothing *logically* wrong

[3]There are some logicians who don't buy this. They think that if X entails Y, there must be some *relevant connection* between X and Y—and there isn't one between $\bot$ and some arbitrary sentence Y. So these logicians develop other, "relevant" logics in which you aren't allowed the explosion rule.

with them (even if they maybe violate all sorts of implicit conversational norms).

Armed with all this, we present the disjunction introduction rule(s):

	⋮	
m	X	
	⋮	
	$X \vee Y$	$\vee$I, m

and

	⋮	
m	X	
	⋮	
	$Y \vee X$	$\vee$I, m

Notice that Y can be *any* sentence whatsoever, so the following is a perfectly acceptable proof:

1	M	
2	$M \vee ([(A \leftrightarrow B) \rightarrow (C \wedge D)] \leftrightarrow [E \wedge F])$	$\vee$I, 1

Using a truth table to show this would have taken 128 lines.

Disjunction Elimination

The disjunction elimination rule also makes use of subproofs.

Suppose that Alice came to the party or Beth came to the party. What can you conclude? Not that Alice came to the party; it might be that Beth came to the party instead. Equally, not that Beth came to the party; for it might be that only Alice came. Disjunctions, just by themselves, are hard to work with.

But suppose that we could somehow show both of the following: first, that Alice coming to the party entails that it was fun: second, that Beth coming to the party entails that it was fun. Then if we know that

Alice or Beth came to the party, then we know that either way, it was fun. This insight can be expressed in the following rule, which is our disjunction elimination (∨E) rule:

m	$X \vee Y$	
i	$\quad X$	
	$\quad \vdots$	
j	$\quad Z$	
k	$\quad Y$	
	$\quad \vdots$	
l	$\quad Z$	
	Z	∨E, m, i–j, k–l

This is obviously a bit clunkier to write down than our previous rules, but the point is fairly simple. Suppose we have some disjunction, $X \vee Y$. Suppose we have two subproofs, showing us that Z follows from the assumption that X, and that Z follows from the assumption that Y. Then we can infer Z itself. As usual, there can be as many lines as you like between i and j, and as many lines as you like between k and l. Moreover, the subproofs and the disjunction can come in any order, and do not have to be adjacent.

Some examples might help illustrate this. Consider this argument:

$$(P \wedge Q) \vee (P \wedge R) \therefore P$$

A proof corresponding to this argument is:

1	$(P \wedge Q) \vee (P \wedge R)$	
2	$\quad P \wedge Q$	
3	$\quad P$	$\wedge$E, 2
4	$\quad P \wedge R$	
5	$\quad P$	$\wedge$E, 4
6	P	$\vee$E, 1, 2–3, 4–5

20.5 Biconditional

We strictly speaking have not got the biconditional '$\leftrightarrow$' in our language of TFL. We have, however, let you use it in symbolizations. We said in section 16.5 that $X \leftrightarrow Y$ is an abbreviation for $(X \rightarrow Y) \wedge (Y \rightarrow X)$.

We then have rules in our natural deduction system that let you exchange $X \leftrightarrow Y$ for its explicit form.

	$\vdots$	
m	$X \leftrightarrow Y$	
	$\vdots$	
	$(X \rightarrow Y) \wedge (Y \leftrightarrow X)$	$\leftrightarrow$E, m

	$\vdots$	
m	$(X \rightarrow Y) \wedge (Y \rightarrow X)$	
	$\vdots$	
	$X \leftrightarrow Y$	$\leftrightarrow$I, m

Practice Exercises (Chapter 20)

I. The following three proofs are missing their citations (rule and line numbers). Add them, to turn them into *bona fide* proofs. Additionally, write down the argument that corresponds to each proof.

1	$P \wedge S$
2	$S \rightarrow R$
3	S
4	R
5	$R \vee E$

1	$A \rightarrow D$	
2	$A \wedge B$	(∗)
3	A	(∗)
4	D	(∗)
5	$D \vee E$	(∗)
6	$(A \wedge B) \rightarrow (D \vee E)$	(∗)

II. For each of the following arguments, give a proof using natural deduction.

a) $J \rightarrow \neg J \therefore \neg J$ (∗)
b) $Q \rightarrow (Q \wedge \neg Q) \therefore \neg Q$ (∗)
c) $A \rightarrow (B \rightarrow C) \therefore (A \wedge B) \rightarrow C$ (∗)
d) $(C \wedge D) \vee E \therefore E \vee D$ (∗)
e) $(Z \wedge K) \vee (K \wedge M), K \rightarrow D \therefore D$ (∗)
f) $P \wedge (Q \vee R), P \rightarrow \neg R \therefore Q \vee E$ (∗)
g) $\neg(P \rightarrow Q) \therefore \neg Q$ (∗)
h) $\neg(P \rightarrow Q) \therefore P$ (∗)

CHAPTER 21

Proofs and Validity

21.1 Proof Strategies

These are all of the basic rules for the proof system for TFL. For ease of reference, they're listed again in appendix 3.7.

There is no simple recipe for proofs, and there is no substitute for practice. Here, though, are some rules of thumb and strategies to keep in mind.

Work backwards from what you want. The ultimate goal is to obtain the conclusion. Look at the conclusion and ask what the introduction rule is for its main logical operator. This gives you an idea of what should happen *just before* the last line of the proof. Then you can treat this line as if it were your goal. Ask what you could do to get to this new goal.

For example: If your conclusion is a conditional $X \to Y$, plan to use the $\to$I rule. This requires starting a subproof in which you assume X. The subproof ought to end with Y. So, what can you do to get Y?

Work forwards from what you have. When you are starting a proof, look at the premises; later, look at the sentences that you have obtained

so far. Think about the elimination rules for the main operators of these sentences. These will often tell you what your options are.

For a short proof, you might be able to eliminate the premises and introduce the conclusion. A long proof is formally just a number of short proofs linked together, so you can fill the gap by alternately working back from the conclusion and forward from the premises.

Try a proof by contradiction. If you cannot find a way to show X directly, try starting by assuming $\neg X$. If a contradiction follows, then you will be able to obtain X by $\neg$E.

Law of Excluded Middle. There's one more convenient approach, closely related to proof by contradiction: if you are allowed to use derived rules, then the rule LEM is sometimes very helpful section 22.1. It will never be necessary, instead a proof by contradiction would do the job instead, but it can be very convenient.

Persist. Try different things. If one approach fails, then try something else.

If the argument is actually valid (which is defined using truth-tables) there will be a proof of it somehow...

Example

Consider $C \therefore B \to ((A \to B) \wedge C)$ How to come up with a proof of this?

1. First step is always to just start organising the proof:

1	C	
	?	
n	$B \to ((A \to B) \wedge C)$	?

2. We'll use introduction rule for the main connective in the conclusion: $\to$Introduction

1	C	
2	$\quad B$	
	$\quad ?$	
$n-1$	$\quad (A \to B) \wedge C$	?
n	$B \to ((A \to B) \wedge C)$	$\to$I, 2–$n-1$

3. To fill out this, we'll use introduction rule for $\wedge$. We already have C, it was our premise, so we just need to show $A \to B$.

1	C	
2	$\quad B$	
	$\quad ?$	
$n-2$	$\quad A \to B$	?
$n-1$	$\quad (A \to B) \wedge C$	$\wedge$I, $n-2$, 1
n	$B \to ((A \to B) \wedge C)$	$\to$I, 2–$n-1$

4. How to get $A \to B$? Introduction rule for $\to$

1	C	
2	$\quad B$	
3	$\quad\quad A$	
	$\quad\quad ?$	
$n-3$	$\quad\quad B$	?
$n-2$	$\quad A \to B$	$\to$I, 3–$n-3$
$n-1$	$\quad (A \to B) \wedge C$	$\wedge$I, $n-2$, 1
n	$B \to ((A \to B) \wedge C)$	$\to$I, 2–$n-1$

5. How can we get to B from what we have? Well we actually already have it! We have alredy assumed B, on line 2 (and that is still

available to us, as we're within that vertical line still). So we can just use Reiteration.

1	C	
2	$\quad B$	
3	$\quad\quad A$	
	$\quad\quad$ no more '?' to fill - we're done	
$n-3$	$\quad\quad B$	R, 2
$n-2$	$\quad A \to B$	$\to$I, 3–$n-3$
$n-1$	$\quad (A \to B) \wedge C$	$\wedge$I, $n-2$, 1
n	$B \to ((A \to B) \wedge C)$	$\to$I, 2–$n-1$

6. Just need to fill out the numbers properly now and we have our completed proof:

1	C	
2	$\quad B$	
3	$\quad\quad A$	
4	$\quad\quad B$	R, 2
5	$\quad A \to B$	$\to$I, 3–4
6	$\quad (A \to B) \wedge C$	$\wedge$I, 5, 1
7	$B \to ((A \to B) \wedge C)$	$\to$I, 2–6

21.2 Proofs and Validity

The system of rules we have set up is not just a game. It helps us understand the validity of arguments.

An argument $X_1, X_2, \ldots, X_n \therefore Y$ may have a proof in the system of natural deduction. Such a proof may look something like:

1	X_1
2	X_2
n	X_n
	⋮
	Y

That is, it will start with the premises as assumptions, and proceed following the rules we have given and finishing with the conclusion. It might also have subproofs along the way, something like:

1	X_1
2	X_2
n	X_n
	⋮
m	Some additional assumption
	⋮
m	Maybe some more assumptions
	⋮
	⋮
	⋮
	Y

But any subproofs need to have been closed by the time we get to Y. So, for example if we gave

1	A	
2	B	
3	B	R, 2

this does not count as a proof corresponding to the argument $A \therefore B$. But

1	A	
2	$\quad B$	
3	$\quad B$	R, 2
4	$B \to B$	→I, 2–3

counts as a proof corresponding to $A \therefore B \to B$ as the subproof has been closed on line 4.

So we have now said when we have a proof in our natural deduction system that corresponds to a particular argument. If we can find a proof then we know that the argument is valid.

> If there is a proof in natural deduction corresponding to the argument $X_1 \dots X_n \therefore Y$, then this argument $X_1 \dots X_n \therefore Y$ is valid.

This property of our proof system is called SOUNDNESS. It holds because we only chose rules that matched valid reasoning steps. Recall that $A \to B, B \therefore A$ is invalid. Had we added a rule such as

	$\vdots$	
n	$X \to Y$	
	$\vdots$	
m	Y	
	$\vdots$	
	X	Do not do this. Does not follow from, n, m

we would then have been able to construct a proof corresponding to the invalid argument $A \to B, B \therefore A$. We do not have such a rule in our system. All the rules we gave in our system will result in proofs of valid arguments.

We can actually strengthen the link between proofs corresponding to arguments and those argument's validity:

> If an argument is valid, then there is a proof of it in natural deduction.

This property of our proof system is called COMPLETENESS.

So for every valid argument there will be some proof. This doesn't mean it is always easy to come up with such a proof, but there will be one. Persist!

CHAPTER 22

Derived rules for TFL

We have introduced all the basic rules of our proof system for TFL. In this section, we will add some additional rules to our system. These will make our system much easier to work with. (However, in chapter 23 we will see that they are not strictly speaking *necessary*.)

22.1 Law of Excluded Middle

We will actually add another rule for how to introduce a disjunction. There are special kinds of disjunctions that don't need further justification: sentences of the form $X \vee \neg X$. In section 10.2 we saw that $A \vee \neg A$ is a tautology: it is true on all valuations. More generally, any sentence of the form $X \vee \neg X$ is a tautology. The rule *Law of Excluded Middle* encodes this fact: it simply says that you are always allowed to write $X \vee \neg X$:

$$\begin{array}{|l} \vdots \\ X \vee \neg X \qquad \text{LEM} \end{array}$$

As always, X can be whatever you want, e.g. $(A \wedge (B \rightarrow C))$. Then the rule tells us, e.g. that you can write $(A \wedge (B \rightarrow C)) \vee \neg (A \wedge (B \rightarrow C))$ on any line of the proof.

Consider the following brain teaser:[1]

> Three people are standing in a row looking at each other.
>
> 
> Alice
>
> 
> Bob
>
> 
> Charlie
>
> Alice is happy. Charlie is not happy. Is there someone who is happy who is looking at someone who is not happy?

...Think about it!
...Answer: Yes. Our Disjunction Elimination rule along with the Law of Excluded Middle allow us to show this. We can demonstrate this in the following argument, which we present in a pseudo-formal style.[2]

1		Bob is either happy or he's not happy
2	|	Suppose Bob is happy
3	|	Then happy Bob is looking at not-happy Charlie
4	|	So someone who is happy is looking at someone who is not
5	|	Suppose Bob is not happy
6	|	Then happy Alice is looking at not-happy Bob
7	|	So someone who is happy is looking at someone who is not
8		Therefore, someone who is happy is looking at someone who is not.

Are you convinced now that someone who is happy is looking at someone who is not happy? If not, find a friend and work through it together. Sometimes it can really help to try walking through the argument together.

[1] Originally by Hector Levesque.

[2] Though, actually, this is most naturally formulated as a validity claim of First Order Logic. We'll walk through the formal proof as formulated in First Order Logic in section 33.2.

Coming up with this sort of argument does just take that moment of inspiration to see how this argument will go (that's why it's a brain teaser). This is often the case with arguments that involve the law of excluded middle. We pick it out of nowhere and have to use our inspiration to see how it might be useful. But hopefully, with more examples you'll become familiar with cases where it might be of use. A strategy that might help is: it's a backup option if everything else fails. If it doesn't look like there's any elimination rules to use on your premises or any introduction rules that can get you to your conclusion, then perhaps LEM is the way forwards.

One more example:

$$P \therefore (P \wedge D) \vee (P \wedge \neg D)$$

Here is a proof corresponding with the argument:

1	P	
2	$D \vee \neg D$	LEM
3	D	
4	$P \wedge D$	$\wedge$I, 1, 3
5	$(P \wedge D) \vee (P \wedge \neg D)$	$\vee$I, 4
6	$\neg D$	
7	$P \wedge \neg D$	$\wedge$I, 1, 6
8	$(P \wedge D) \vee (P \wedge \neg D)$	$\vee$I, 7
9	$(P \wedge D) \vee (P \wedge \neg D)$	$\vee$E, 2, 3–5, 6–8

22.2 Disjunctive syllogism

Here is a very natural argument form.

> Elizabeth is in Massachusetts or in DC. She is not in DC.
> So, she is in Massachusetts.

This inference pattern is called *disjunctive syllogism*. We add it to our proof system as follows:

m	$X \vee Y$	
n	$\neg X$	
	Y	DS, m, n

and

m	$X \vee Y$	
n	$\neg Y$	
	X	DS, m, n

As usual, the disjunction and the negation of one disjunct may occur in either order and need not be adjacent.

22.3 Modus tollens

Another useful pattern of inference is embodied in the following argument:

> If Mitt has won the election, then he is in the White House. He is not in the White House. So he has not won the election.

This inference pattern is called *modus tollens*. The corresponding rule is:

m	$X \to Y$	
n	$\neg Y$	
	$\neg X$	MT, m, n

As usual, the premises may occur in either order.

22.4 Double-negation

A sentence $\neg\neg X$ is always logically equivalent to X. We can add rules to our system that encode this idea: allowing us to immediately eliminate or introduce double negations:

m	$\neg\neg X$	
	X	DNE, m

m	X	
	$\neg\neg X$	DNI, m

That said, you should be aware that in ordinary language we can sometimes speak in a way that is similar to, but not quite, a double negation. Consider: 'Jane is not *un*happy'. Arguably, one cannot infer 'Jane is happy'. Perhaps the speaker is using this unusual indirect phrasing to draw attention to the possible difference between 'unhappy' and 'not unhappy'. Perhaps what they mean to suggest is that 'Jane is in a state of profound indifference'. Here, then, 'Jane is unhappy' should not be thought of as equivalent to 'It is not the case that Jane is happy', and it should not be symbolised as $\neg H$ but should rather be a separate atomic sentence. So 'Jane is not unhappy' is not then seen as a double negation.

22.5 Explosion

From a contradiction anything follows. This is called the rule of explosion. It is also sometimes called *ex falso quod libet*.

m	X	
n	$\neg X$	
	Y	Explosion, m, n

22.6 De Morgan Rules

Our final additional rules are called De Morgan's Laws. (These are named after Augustus De Morgan.) The shape of the rules should be familiar from truth tables.

The first De Morgan rule is:

m	$\neg(X \wedge Y)$	
	$\neg X \vee \neg Y$	DeM, m

The second De Morgan is the reverse of the first:

m	$\neg X \vee \neg Y$	
	$\neg(X \wedge Y)$	DeM, m

The third De Morgan rule is the *dual* of the first:

m	$\neg(X \vee Y)$	
	$\neg X \wedge \neg Y$	DeM, m

And the fourth is the reverse of the third:

m	$\neg X \wedge \neg Y$	
	$\neg(X \vee Y)$	DeM, m

There are many more rules one could add to the system as derived rules. But these are all the ones we'll introduce.

Practice Exercises (Chapter 22)

I. The following proofs are missing their citations (rule and line numbers). Add them wherever they are required:

1	$W \to \neg B$	
2	$A \wedge W$	
3	$B \vee (J \wedge K)$	
4	W	
5	$\neg B$	
6	$J \wedge K$	
7	K	

1	$\neg O \to L$	
2	$L \vee \neg O$	
3		$\neg L$
4		$\neg O$
5		L
6	$\neg\neg L$	
7	L	

1	$Z \to (C \wedge \neg N)$	
2	$\neg Z \to (N \wedge \neg C)$	
3	$Z \vee \neg Z$	
4		Z
5		$C \wedge \neg N$
6		C
7		$N \vee C$
8	$Z \to (N \vee C)$	
9		$\neg Z$
10		$N \wedge \neg C$
11		N
12		$N \vee C$
13	$\neg Z \to (N \vee C)$	
14	$N \vee C$	

II. Give a proof for each of these arguments:

1. $\neg F \to G, F \to H \therefore G \vee H$ (*)
2. $E \vee F$, $F \vee G$, $\neg F \therefore E \wedge G$
3. $M \vee (N \to M) \therefore \neg M \to \neg N$
4. $(M \vee N) \wedge (O \vee P)$, $N \to P$, $\neg P \therefore M \wedge O$
5. $A \wedge B) \vee (A \wedge C)$, $\neg(A \wedge D)$, $D \vee M \therefore M$

CHAPTER 23

Derivations of the derived rules

In this section, we will see why we introduced the rules of our proof system in two separate batches. In particular, we want to show that the additional rules of chapter 22 are not strictly speaking necessary, but can be derived from the basic rules of chapter 19.

23.1 Derivation of Disjunctive syllogism

Suppose that you are in a proof, and you have something of this form:

m	$X \vee Y$
n	$\neg X$

You now want, on line k, to prove Y. You can do this with the rule of DS, introduced in chapter 22, but equally well, you can do this with the *basic* rules of chapter 19:

m	$X \vee Y$	
n	$\neg X$	
k	| X	
$k+1$	| | $\neg Y$	
$k+2$	| | $X \wedge \neg X$	$\wedge$I, k, n
$k+3$	| Y	$\neg$E, $k+1$–$k+2$
$k+4$	| Y	
$k+5$	| Y	R, $k+4$
$k+6$	Y	$\vee$E, m, k–$k+3$, $k+4$–$k+5$

To be clear: this is not a proof. Rather, it is a proof *scheme*. (This is why we use letters like m and k to label the lines of the proof rather than numbers.) Whatever sentences of TFL we plugged in for 'X' or 'Y', and whatever lines we were working on, we could produce a bona fide proof. So you can think of this as a recipe for producing proofs.

Indeed, it is a recipe which shows us that, anything we can prove using the rule DS, we can prove (with a few more lines) using just the other rules of chapter 19.

23.2 Derivation of Modus Tollens

Suppose in the course of you proof you already have $X \rightarrow Y$, say on line m, and $\neg Y$ on line n. At some later line, k, you want to get $\neg X$. You can do this with the rule of Modus Tollens (MT), introduced in chapter 22. But you could also do this with the *basic* rules of chapter 19:

m	$X \rightarrow Y$	
n	$\neg Y$	
k	| X	
$k+1$	| Y	$\rightarrow$E, m, k
$k+2$	| $Y \wedge \neg Y$	$\wedge$I, $k+1$, n
$k+3$	$\neg X$	$\neg$I, k–$k+2$

Again, the rule of MT can be derived from the *basic* rules of chapter 19.

23.3 Derivation of Double-negation rules

Consider the following deduction schema:

m	X	
j	$\quad\vdots$	
$j+1$	$\quad\neg X$	
	$\quad\overline{X \wedge \neg X}$	$\wedge$I, m, j
$j+2$	$\neg\neg X$	$\neg$I, j–$j+1$

and

m	$\neg\neg X$	
j	$\quad\vdots$	
$j+1$	$\quad\neg X$	
	$\quad\overline{\neg X \wedge \neg\neg X}$	$\wedge$I, m, j
$j+2$	X	$\neg$E, j–$j+1$

So again, we can derive the double negations rules from the *basic* rules of chapter 19.

23.4 Derivation of Explosion

Here is a demonstration of how we could derive explosion using the basic rules:

m	X	
n	$\neg X$	
$n+1$	$\quad\neg Y$	
	$\quad\overline{X \wedge \neg X}$	$\wedge$I, m, n
$n+2$	Y	$\neg$E, $n+1$–$n+2$

23.5 Derivation of De Morgan rules

Here is a demonstration of how we could derive the first De Morgan rule:

m	$\neg(X \wedge Y)$	
j	$X \vee \neg X$	LEM
k	$\quad X$	
$k+1$	$\quad\quad Y$	
$k+2$	$\quad\quad X \wedge Y$	$\wedge$I, k, $k+1$
$k+3$	$\quad\quad (X \wedge Y) \wedge \neg(X \wedge Y)$	$\wedge$I, $k+2$, m
$k+4$	$\quad \neg Y$	$\neg$I, $k+1$–$k+3$
$k+5$	$\quad \neg X \vee \neg Y$	$\vee$I, $k+4$
$k+6$	$\quad \neg X$	
$k+7$	$\quad \neg X \vee \neg Y$	$\vee$I, $k+6$
$k+8$	$\neg X \vee \neg Y$	$\vee$E, j, k–$k+5$, $k+6$–$k+7$

Here is a demonstration of how we could derive the second De Morgan rule:

m	$\neg X \vee \neg Y$	
$m+1$	| $\neg X$	
k	| | $X \wedge Y$	
$k+1$	| | X	$\wedge$E, k
$k+2$	| | $\neg X$	R, $m+1$
$k+3$	| $\neg(X \wedge Y)$	$\neg$I, k–$k+2$
$k+4$	| $\neg Y$	
l	| | $X \wedge Y$	
$l+1$	| | Y	$\wedge$E, k
$l+2$	| | $Y \wedge \neg Y$	$\wedge$I, $l+1$, $k+4$
$l+3$	| $\neg(X \wedge Y)$	$\neg$I, l–$l+2$
$l+4$	$\neg(X \wedge Y)$	$\vee$E, m, $m+1$–$k+3$, $k+4$–$l+3$

Similar demonstrations can be offered explaining how we could derive the third and fourth De Morgan rules. These are left as exercises.

Practice Exercises (Chapter 23)

I. Provide proof schemes that justify the addition of the third and fourth De Morgan rules as derived rules.

II. The proofs you offered in response to the practice exercises of chapter 22–25 used derived rules. Replace the use of derived rules, in such proofs, with only basic rules. You will find some 'repetition' in the resulting proofs; in such cases, offer a streamlined proof using only basic rules. (This will give you a sense, both of the power of derived rules, and of how all the rules interact.)

CHAPTER 24

Soundness and completeness

(Non-examinable)A very important result:

> A TFL argument is valid if and only if it can be given a proof in this natural deduction system.

In this chapter, we explain a bit more about how such an argument would go.

Logical Consequence

For this chapter we will make use of a further symbols:

We use the symbol $\vDash$ as shorthand for logical consequence, that is, instead of saying that the TFL-sentence Y. is a logical consequence of the TFL sentences $X_1, X_2, \ldots$ and X_n, we will abbreviate this by:

$$X_1, X_2, \ldots, X_n \vDash Y$$

The symbol '$\vDash$' is known as *the double-turnstile*, since it looks like a turnstile with two horizontal beams.

Let us be clear be clear. '$\models$' is not a symbol of TFL. Rather, it is a symbol of our metalanguage, augmented English (recall the difference between object language and metalanguage from chapter 6). So the metalanguage sentence:

- $P, P \rightarrow Q \models Q$

is just an abbreviation of the sentence:

- The TFL sentences 'Q' is a logical consequence of 'P' and '$P \rightarrow Q$'.

Note that there is no limit on the number of TFL sentences that can be mentioned before the symbol '$\models$'. Indeed, we can even consider the limiting case where there is no sentence (this should ring familiar from **??**):

$$\models Y$$

Proof

The following expression:

$$X_1, X_2, \ldots, X_n \vdash Y$$

means that there is some proof which starts with assumptions among $X_1, X_2, \ldots, X_n$ and ends with Y (and contains no undischarged assumptions other than those we started with). Derivatively, we will write:

$$\vdash X$$

to mean that there is a proof of X with no assumptions.

The symbol '$\vdash$' is called the *single turnstile*. We want to emphasize that this is not the double turnstile symbol ('$\models$') that we introduced for 'logical consequence'. The single turnstile, '$\vdash$', concerns the existence of proofs; the double turnstile, '$\models$', concerns the existence of valuations (or interpretations, when used for FOL). *They are very different notions.*

24.1 Their equivalence

However, it turns out that they are equivalent. That is:

$$X_1, X_2, \ldots, X_n \vdash Y$$
if and only if
$$X_1, X_2, \ldots, X_n \models Y$$

A full proof here goes well beyond the scope of this book. However, we can sketch what it would be like.

Soundness

This argument from $\vdash$ to $\models$ is the problem of SOUNDNESS. A proof system is SOUND if there are no derivations of arguments that can be shown invalid by truth tables. Demonstrating that the proof system is sound would require showing that *any* possible proof is the proof of a valid argument. It would not be enough simply to succeed when trying to prove many valid arguments and to fail when trying to prove invalid ones.

The proof that we will sketch depends on the fact that we initially defined a sentence of TFL using a inductive definition (as in chapter 7). We could have also used inductive definitions to define a proper proof in TFL and a proper truth table. (Although we didn't.) If we had these definitions, we could then use a *inductive proof* to show the soundness of TFL. A inductive proof works the same way as a inductive definition. With the inductive definition, we identified a group of base elements that were stipulated to be examples of the thing we were trying to define. In the case of a TFL sentence, the base class was the set of sentence letters A, B, C, ... We just announced that these were sentences. The second step of a inductive definition is to say that anything that is built up from your base class using certain rules also counts as an example of the thing you are defining. In the case of a definition of a sentence, the rules corresponded to the five sentential connectives (as in chapter 7). Once you have established a inductive definition, you can use that definition to show that all the members of the class you have defined have a certain property. You simply prove that the property is true of the members of the base class, and then you prove that the rules for extending the base class don't change the property. This is what it means to give a inductive proof.

Even though we don't have a inductive definition of a proof in TFL, we can sketch how a inductive proof of the soundness of TFL would go. Imagine a base class of one-line proofs, one for each of our basic rules of inference. The members of this class would look like this $X, Y \vdash X \wedge Y$; $X \wedge Y \vdash X$; $X \vee Y, \neg X \vdash Y$ ... etc. Since some rules have a couple different forms, we would have to have add some members to this base class, for instance $X \wedge Y \vdash Y$ Notice that these are all statements in the metalanguage. The proof that TFL is sound is not a part of TFL, because TFL does not have the power to talk about itself.

You can use truth tables to prove to yourself that each of these one-line proofs in this base class is valid$_{\vDash}$. For instance the proof $X, Y \vdash X \wedge Y$ corresponds to a truth table that shows $X, Y \vDash X \wedge Y$ This establishes the first part of our inductive proof.

The next step is to show that adding lines to any proof will never change a valid$_{\vDash}$ proof into an invalid$_{\vDash}$ one. We would need to do this for each of our basic rules of inference. So, for instance, for $\wedge$I we need to show that for any proof $X_1, \ldots, X_n \vdash Y$ adding a line where we use $\wedge$I to infer $Z \wedge V$, where $Z \wedge V$ can be legitimately inferred from $X_1, \ldots, X_n$, Y, would not change a valid proof into an invalid proof. But wait, if we can legitimately derive $Z \wedge V$ from these premises, then Z and V must be already available in the proof. They are either already among $X_1, \ldots, X_n$, B, or can be legitimately derived from them. As such, any truth table line in which the premises are true must be a truth table line in which Z and V are true. According to the characteristic truth table for $\wedge$, this means that $Z \wedge V$ is also true on that line. Therefore, $Z \wedge V$ validly follows from the premises. This means that using the $\wedge$E rule to extend a valid proof produces another valid proof.

In order to show that the proof system is sound, we would need to show this for the other inference rules. Since the derived rules are consequences of the basic rules, it would suffice to provide similar arguments for the other basic rules. This tedious exercise falls beyond the scope of this book.

So we have shown that $X \vdash Y$ implies $X \vDash Y$. What about the other direction, that is why think that *every* argument that can be shown valid using truth tables can also be proven using a derivation.

Completeness

This is the problem of completeness. A proof system has the property of COMPLETENESS if and only if there is a derivation of every semantically valid argument. Proving that a system is complete is generally harder than proving that it is sound. Proving that a system is sound amounts to showing that all of the rules of your proof system work the way they are supposed to. Showing that a system is complete means showing that you have included *all* the rules you need, that you haven't left any out. Showing this is beyond the scope of this book. The important point is that, happily, the proof system for TFL is both sound and complete. This is not the case for all proof systems or all formal languages. Because it is true of TFL, we can choose to give proofs or give truth tables—whichever is easier for the task at hand.

24.2 Other Semantic and Proof Theoretic Notions

Now we know that the proof theoretic and truth-table methods are equivalent, we can use them interchangable depending on which is more useful.

We can also use proof theoretic methods for determining other logical notions, such as being consistent. We summarise how one would define them in table 24.1

In fact, we can give general guidelines about when it's best to give proofs and when it is best to give truth tables. We do this in table 24.2:

Practice Exercises (Chapter 24)

I. Use either a derivation or a truth table for each of the following.

a) Show that $A \rightarrow (((B \wedge C) \vee D) \rightarrow A)$ is a tautology.
b) Show that $A \rightarrow (A \rightarrow B)$ is not a tautology
c) Show that the sentence $A \rightarrow \neg A$ is not a contradiction.
d) Show that the sentence $A \leftrightarrow \neg A$ is a contradiction.
e) Show that the sentence $\neg(W \rightarrow (J \vee J))$ is contingent
f) Show that the sentence $\neg(X \vee (Y \vee Z)) \vee (X \vee (Y \vee Z))$ is not contingent
g) Show that the sentence $B \rightarrow \neg S$ is equivalent to the sentence $\neg\neg B \rightarrow \neg S$
h) Show that the sentence $\neg(X \vee O)$ is not equivalent to the sentence $X \wedge O$
i) Show that the sentences $\neg(A \vee B)$, C, $C \rightarrow A$ are jointly inconsistent.
j) Show that the sentences $\neg(A \vee B)$, $\neg B$, $B \rightarrow A$ are jointly consistent
k) Show that $\neg(A \vee (B \vee C)) \therefore \neg C$ is valid.
l) Show that $\neg(A \wedge (B \vee C)) \therefore \neg C$ is invalid.

II. Use either a derivation or a truth table for each of the following.

a) Show that $A \rightarrow (B \rightarrow A)$ is a tautology
b) Show that $\neg(((N \leftrightarrow Q) \vee Q) \vee N)$ is not a tautology
c) Show that $Z \vee (\neg Z \leftrightarrow Z)$ is contingent
d) show that $(L \leftrightarrow ((N \rightarrow N) \rightarrow L)) \vee H$ is not contingent
e) Show that $(A \leftrightarrow A) \wedge (B \wedge \neg B)$ is a contradiction
f) Show that $(B \leftrightarrow (C \vee B))$ is not a contradiction.
g) Show that $((\neg X \leftrightarrow X) \vee X)$ is equivalent to X

Concept	Truth table (semantic) definition	Proof-theoretic (syntactic) definition
Tautology	A sentence whose truth table only has Ts under the main connective	A sentence that can be derived without any premises.
Contradiction	A sentence whose truth table only has Fs under the main connective	A sentence whose negation can be derived without any premises
Contingent sentence	A sentence whose truth table contains both Ts and Fs under the main connective	A sentence that is not a theorem or contradiction
Equivalent sentences	The columns under the main connectives are identical.	The sentences can be derived from each other
Inconsistent sentences	Sentences which do not have a single line in their truth table where they are all true.	Sentences from which one can derive a contradiction
Consistent sentences	Sentences which have at least one line in their truth table where they are all true.	Sentences which are not inconsistent
Valid argument	An argument whose truth table has no lines where there are all Ts under main connectives for the premises and an F under the main connective for the conclusion.	An argument where one can derive the conclusion from the premises

Table 24.1: Two ways to define logical concepts.

Logical property	**To prove it present**	**To prove it absent**
Being a tautology	Derive the sentence	Find the false line in the truth table for the sentence
Being a contradiction	Derive the negation of the sentence	Find the true line in the truth table for the sentence
Contingency	Find a false line and a true line in the truth table for the sentence	Prove the sentence or its negation
Equivalence	Derive each sentence from the other	Find a line in the truth tables for the sentence where they have different values
Consistency	Find a line in truth table for the sentence where they all are true	Derive a contradiction from the sentences
Validity	Derive the conclusion from the premises	Find no line in the truth table where the premises are true and the conclusion false.

Table 24.2: When to provide a truth table and when to provide a proof.

h) Show that $F \wedge (K \wedge R)$ is not equivalent to $(F \leftrightarrow (K \leftrightarrow R))$
i) Show that the sentences $\neg(W \rightarrow W)$, $(W \leftrightarrow W) \wedge W$, $E \vee (W \rightarrow \neg(E \wedge W))$ are inconsistent.
j) Show that the sentences $\neg R \vee C$, $(C \wedge R) \rightarrow \neg R$, $(\neg(R \vee R) \rightarrow R)$ are consistent.
k) Show that $\neg\neg(C \leftrightarrow \neg C), ((G \vee C) \vee G) \therefore ((G \rightarrow C) \wedge G)$ is valid.
l) Show that $\neg\neg L, (C \rightarrow \neg L) \rightarrow C) \therefore \neg C$ is invalid.

CHAPTER 25

Proof-theoretic concepts

Armed with our '$\vdash$' symbol, we can introduce some new terminology.

Definition

X is a THEOREM iff $\vdash X$

To illustrate this, suppose we want to prove that '$\neg(A \wedge \neg A)$' is a theorem. So we must start our proof without *any* assumptions. However, since we want to prove a sentence whose main logical operator is a negation, we will want to immediately begin a subproof, with the additional assumption '$A \wedge \neg A$', and show that this leads to contradiction. All told, then, the proof looks like this:

1	$A \wedge \neg A$	
2	A	$\wedge$E, 1
3	$\neg A$	$\wedge$E, 1
4	$\neg(A \wedge \neg A)$	$\neg$I, 1–3

We have therefore proved '$\neg(A \wedge \neg A)$' on no (undischarged) assumptions. This particular theorem is an instance of what is sometimes called *the Law of Non-Contradiction*.

To show that something is a theorem, you just have to find a suitable proof. It is typically much harder to show that something is *not*

a theorem. To do this, you would have to demonstrate, not just that certain proof strategies fail, but that *no* proof is possible. Even if you fail in trying to prove a sentence in a thousand different ways, perhaps the proof is just too long and complex for you to make out. Perhaps you just didn't try hard enough.

Here is another new bit of terminology:

Definition

Two sentences X and Y are PROVABLY EQUIVALENT iff each can be proved from the other; i.e., both $X \vdash Y$ and $Y \vdash X$.

As in the case of showing that a sentence is a theorem, it is relatively easy to show that two sentences are provably equivalent: it just requires a pair of proofs. Showing that sentences are *not* provably equivalent would be much harder: it is just as hard as showing that a sentence is not a theorem.

Here is a third, related, bit of terminology:

Definition

The sentences $X_1, X_2, \ldots, X_n$ are PROVABLY INCONSISTENT iff a contradiction can be proved from them, i.e. $X_1, X_2, \ldots, X_n \vdash Y \wedge \neg Y$ for some Y. If they are not INCONSISTENT, we call them PROVABLY CONSISTENT.

It is easy to show that some sentences are provably inconsistent: you just need to prove a contradiction from assuming all the sentences. Showing that some sentences are not provably inconsistent is much harder. It would require more than just providing a proof or two; it would require showing that no proof of a certain kind is *possible*.

This table summarises whether one or two proofs suffice, or whether we must reason about all possible proofs.

	Yes	**No**
theorem?	one proof	all possible proofs
inconsistent?	one proof	all possible proofs
equivalent?	two proofs	all possible proofs
consistent?	all possible proofs	one proof

Practice Exercises (Chapter 25)

I. Show that each of the following sentences is a theorem:

a) $O \rightarrow O$
b) $N \vee \neg N$
c) $J \leftrightarrow (J \vee (L \wedge \neg L))$
d) $((A \rightarrow B) \rightarrow A) \rightarrow A$

II. Provide proofs to show each of the following:

a) $C \rightarrow (E \wedge G), \neg C \rightarrow G \vdash G$
b) $M \wedge (\neg N \rightarrow \neg M) \vdash (N \wedge M) \vee \neg M$
c) $(Z \wedge K) \leftrightarrow (Y \wedge M), D \wedge (D \rightarrow M) \vdash Y \rightarrow Z$
d) $(W \vee X) \vee (Y \vee Z), X \rightarrow Y, \neg Z \vdash W \vee Y$

III. Show that each of the following pairs of sentences are provably equivalent:

a) $R \leftrightarrow E$, $E \leftrightarrow R$
b) G, $\neg\neg\neg\neg G$
c) $T \rightarrow S$, $\neg S \rightarrow \neg T$
d) $U \rightarrow I$, $\neg(U \wedge \neg I)$
e) $\neg(C \rightarrow D), C \wedge \neg D$
f) $\neg G \leftrightarrow H$, $\neg(G \leftrightarrow H)$

IV. If you know that $X \vdash Y$, what can you say about $(X \wedge Z) \vdash Y$? What about $(X \vee Z) \vdash Y$? Explain your answers.

V. In this chapter, we claimed that it is just as hard to show that two sentences are not provably equivalent, as it is to show that a sentence is not a theorem. Why did we claim this? (*Hint*: think of a sentence that would be a theorem iff X and Y were provably equivalent.)

APPENDIX A

Answers to practice exercises

Answers to Practice Exercises from Chapter 19

I. The following proofs are missing their citations (rule and line numbers). Add them, to turn them into *bona fide* proofs. Additionally, write down the argument that corresponds to each proof.

1	$P \wedge S$	
2	$S \rightarrow R$	
3	S	$\wedge$E, 1
4	R	$\rightarrow$E, 2, 3
5	$A \vee E$	$\vee$I, 4

II. Give a proof corresponding to each of the following arguments:

1. $A \wedge (B \vee C) \therefore (B \vee C) \wedge A$

1	$A \wedge (B \vee C)$	
2	A	$\wedge$E, 1
3	$B \vee C$	$\wedge$E, 1
4	$(B \vee C) \wedge A$	$\wedge$I, 3, 2

2. $A \rightarrow B,\ A \rightarrow C,\ A \wedge D \ \therefore\ B \wedge C$

Answers to Practice Exercises from Chapter 20

I. The following three proofs are missing their citations (rule and line numbers). Add them, to turn them into *bona fide* proofs. Additionally, write down the argument that corresponds to each proof.

1	$P \wedge S$	
2	$S \rightarrow R$	
3	S	$\wedge$E, 1
4	R	$\rightarrow$E, 2, 3
5	$R \vee E$	$\vee$I, 4

1	$A \rightarrow D$	
2	$\quad A \wedge B$	
3	$\quad A$	$\wedge$E, 2
4	$\quad D$	$\rightarrow$E, 1, 3
5	$\quad D \vee E$	$\vee$I, 4
6	$(A \wedge B) \rightarrow (D \vee E)$	$\rightarrow$I, 2–5

II. For each of the following arguments, give a proof using natural deduction.

a) $J \rightarrow \neg J \therefore \neg J$

1	$J \rightarrow \neg J$	
2	$\quad J$	
3	$\quad \neg J$	$\rightarrow$E, 1, 2
4	$\quad J \wedge \neg J$	$\wedge$I, 2, 3
5	$\neg J$	$\neg$I, 2–4

b) $Q \to (Q \wedge \neg Q) \therefore \neg Q$

1	$Q \to (Q \wedge \neg Q)$	
2	$\quad Q$	
3	$\quad Q \wedge \neg Q$	$\to$E, 1, 2
4	$\quad \neg Q$	$\wedge$E, 3
5	$\quad Q \wedge \neg Q$	$\wedge$I, 2, 4
6	$\neg Q$	$\neg$I, 2–5

c) $A \to (B \to C) \therefore (A \wedge B) \to C$

1	$A \to (B \to C)$	
2	$\quad A \wedge B$	
3	$\quad A$	$\wedge$E, 2
4	$\quad B$	$\wedge$E, 2
5	$\quad B \to C$	$\to$E, 1, 3
6	$\quad C$	$\to$E, 5, 4
7	$(A \wedge B) \to C$	$\to$I, 2–6

d) $(C \wedge D) \vee E \therefore E \vee D$

1	$(C \wedge D) \vee E$	
2	$\quad C \wedge D$	
3	$\quad D$	$\wedge$E, 2
4	$\quad E \vee D$	$\vee$I, 3
5	$\quad E$	
6	$\quad E \vee D$	$\vee$I, 5
7	$E \vee D$	$\vee$E, 1, 2–4, 5–6

e) $(Z \wedge K) \vee (K \wedge M), K \to D \therefore D$

1	$(Z \wedge K) \vee (K \wedge M)$	
2	$K \rightarrow D$	
3	$\quad Z \wedge K$	
4	$\quad K$	$\wedge$E, 3
5	$\quad D$	$\rightarrow$E, 2, 4
6	$\quad K \wedge M$	
7	$\quad K$	$\wedge$E, 6
8	$\quad D$	$\rightarrow$E, 2, 7
9	D	$\vee$E, 1, 3–5, 6–8

f) $P \wedge (Q \vee R), P \rightarrow \neg R \therefore Q \vee E$

1	$P \wedge (Q \vee R)$	
2	$P \rightarrow \neg R$	
3	P	$\wedge$E, 1
4	$Q \vee R$	$\wedge$E, 1
5	$\quad Q$	
6	$\quad Q \vee E$	$\vee$I, 5
7	$\quad R$	
8	$\quad \neg R$	$\rightarrow$E, 2, 3
9	$\quad\quad \neg(Q \vee E)$	
10	$\quad\quad R \wedge \neg R$	$\wedge$I, 7, 8
11	$\quad Q \vee E$	$\neg$E, 9–10
12	$Q \vee E$	$\vee$E, 4, 5–6, 7–11

g) $\neg(P \rightarrow Q) \therefore \neg Q$

1	$\neg(P \to Q)$	
2	$\quad Q$	
3	$\quad\quad P$	
4	$\quad\quad Q$	R, 2
5	$\quad P \to Q$	$\to$I, 3–4
6	$\quad (P \to Q) \wedge \neg(P \to Q)$	$\wedge$I, 5, 1
7	$\neg Q$	$\neg$I, 2–6

h) $\neg(P \to Q) \therefore P$

1	$\neg(P \to Q)$	
2	$\quad \neg P$	
3	$\quad\quad P$	
4	$\quad\quad\quad \neg Q$	
5	$\quad\quad\quad P \wedge \neg P$	$\wedge$I, 3, 2
6	$\quad\quad Q$	$\neg$E, 4–5
7	$\quad P \to Q$	$\to$I, 3–6
8	$\quad (P \to Q) \wedge \neg(P \to Q)$	$\wedge$I, 7, 1
9	P	$\neg$E, 2–8

Answers to Practice Exercises from Chapter 22

I. The following proofs are missing their citations (rule and line numbers). Add them wherever they are required:

1	$W \to \neg B$
2	$A \wedge W$
3	$B \vee (J \wedge K)$
4	W
5	$\neg B$
6	$J \wedge K$
7	K

1	$\neg O \to L$
2	$L \vee \neg O$
3	$\quad \neg L$
4	$\quad \neg O$
5	$\quad L$
6	$\neg\neg L$
7	L

1	$Z \to (C \wedge \neg N)$
2	$\neg Z \to (N \wedge \neg C)$
3	$Z \vee \neg Z$
4	$\quad Z$
5	$\quad C \wedge \neg N$
6	$\quad C$
7	$\quad N \vee C$
8	$Z \to (N \vee C)$
9	$\quad \neg Z$
10	$\quad N \wedge \neg C$
11	$\quad N$
12	$\quad N \vee C$
13	$\neg Z \to (N \vee C)$
14	$N \vee C$

II. Give a proof for each of these arguments:

1. $\neg F \to G, F \to H \therefore G \vee H$

1	$\neg F \to G$	
2	$F \to H$	
3	$F \vee \neg F$	LEM
4	$\quad F$	
5	$\quad H$	$\to$E, 2, 4
6	$\quad G \vee H$	$\vee$I, 5
7	$\quad \neg F$	
8	$\quad G$	$\to$E, 1, 7
9	$\quad G \vee H$	$\vee$I, 8
10	$G \vee H$	$\vee$E, 3, 4–6, 7–9

2. $E \vee F, F \vee G, \neg F \therefore E \wedge G$
3. $M \vee (N \to M) \therefore \neg M \to \neg N$
4. $(M \vee N) \wedge (O \vee P), N \to P, \neg P \therefore M \wedge O$
5. $A \wedge B) \vee (A \wedge C), \neg(A \wedge D), D \vee M \therefore M$

Answers to Practice Exercises from Chapter 23

I. Provide proof schemes that justify the addition of the third and fourth De Morgan rules as derived rules.

II. The proofs you offered in response to the practice exercises of chapter 22–25 used derived rules. Replace the use of derived rules, in such proofs, with only basic rules. You will find some 'repetition' in the resulting proofs; in such cases, offer a streamlined proof using only basic rules. (This will give you a sense, both of the power of derived rules, and of how all the rules interact.)

Answers to Practice Exercises from Chapter 24

I. Use either a derivation or a truth table for each of the following.

a) Show that $A \rightarrow (((B \wedge C) \vee D) \rightarrow A)$ is a tautology.
b) Show that $A \rightarrow (A \rightarrow B)$ is not a tautology
c) Show that the sentence $A \rightarrow \neg A$ is not a contradiction.
d) Show that the sentence $A \leftrightarrow \neg A$ is a contradiction.
e) Show that the sentence $\neg(W \rightarrow (J \vee J))$ is contingent
f) Show that the sentence $\neg(X \vee (Y \vee Z)) \vee (X \vee (Y \vee Z))$ is not contingent
g) Show that the sentence $B \rightarrow \neg S$ is equivalent to the sentence $\neg\neg B \rightarrow \neg S$
h) Show that the sentence $\neg(X \vee O)$ is not equivalent to the sentence $X \wedge O$
i) Show that the sentences $\neg(A \vee B)$, C, $C \rightarrow A$ are jointly inconsistent.
j) Show that the sentences $\neg(A \vee B)$, $\neg B$, $B \rightarrow A$ are jointly consistent
k) Show that $\neg(A \vee (B \vee C)) \therefore \neg C$ is valid.
l) Show that $\neg(A \wedge (B \vee C)) \therefore \neg C$ is invalid.

II. Use either a derivation or a truth table for each of the following.

a) Show that $A \rightarrow (B \rightarrow A)$ is a tautology
b) Show that $\neg(((N \leftrightarrow Q) \vee Q) \vee N)$ is not a tautology
c) Show that $Z \vee (\neg Z \leftrightarrow Z)$ is contingent
d) show that $(L \leftrightarrow ((N \rightarrow N) \rightarrow L)) \vee H$ is not contingent
e) Show that $(A \leftrightarrow A) \wedge (B \wedge \neg B)$ is a contradiction
f) Show that $(B \leftrightarrow (C \vee B))$ is not a contradiction.
g) Show that $((\neg X \leftrightarrow X) \vee X)$ is equivalent to X
h) Show that $F \wedge (K \wedge R)$ is not equivalent to $(F \leftrightarrow (K \leftrightarrow R))$
i) Show that the sentences $\neg(W \rightarrow W)$, $(W \leftrightarrow W) \wedge W$, $E \vee (W \rightarrow \neg(E \wedge W))$ are inconsistent.
j) Show that the sentences $\neg R \vee C$, $(C \wedge R) \rightarrow \neg R$, $(\neg(R \vee R) \rightarrow R)$ are consistent.
k) Show that $\neg\neg(C \leftrightarrow \neg C), ((G \vee C) \vee G) \therefore ((G \rightarrow C) \wedge G)$ is valid.
l) Show that $\neg\neg L, (C \rightarrow \neg L) \rightarrow C) \therefore \neg C$ is invalid.

Answers to Practice Exercises from Chapter 25

I. Show that each of the following sentences is a theorem:

a) $O \rightarrow O$
b) $N \vee \neg N$

c) $J \leftrightarrow (J \vee (L \wedge \neg L))$
d) $((A \rightarrow B) \rightarrow A) \rightarrow A$

II. Provide proofs to show each of the following:

a) $C \rightarrow (E \wedge G), \neg C \rightarrow G \vdash G$
b) $M \wedge (\neg N \rightarrow \neg M) \vdash (N \wedge M) \vee \neg M$
c) $(Z \wedge K) \leftrightarrow (Y \wedge M), D \wedge (D \rightarrow M) \vdash Y \rightarrow Z$
d) $(W \vee X) \vee (Y \vee Z), X \rightarrow Y, \neg Z \vdash W \vee Y$

III. Show that each of the following pairs of sentences are provably equivalent:

a) $R \leftrightarrow E$, $E \leftrightarrow R$
b) G, $\neg\neg\neg\neg G$
c) $T \rightarrow S$, $\neg S \rightarrow \neg T$
d) $U \rightarrow I$, $\neg(U \wedge \neg I)$
e) $\neg(C \rightarrow D), C \wedge \neg D$
f) $\neg G \leftrightarrow H$, $\neg(G \leftrightarrow H)$

IV. If you know that $X \vdash Y$, what can you say about $(X \wedge Z) \vdash Y$? What about $(X \vee Z) \vdash Y$? Explain your answers.

V. In this chapter, we claimed that it is just as hard to show that two sentences are not provably equivalent, as it is to show that a sentence is not a theorem. Why did we claim this? (*Hint*: think of a sentence that would be a theorem iff X and Y were provably equivalent.)